COMP102 Lecture 9: Graphs II: Algorithms

BFS, DFS, components, and shortest paths

Jalal Maaouni
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UM6P University Mohammed VI Polytechnic
THE UM6P VANGUARD CENTER
#Empowering Minds.

Plan

1. Motivation
2. Recap adjacency lists
3. Traversal problem
4. BFS
5. Shortest path intuition
6. Path reconstruction
7. DFS
8. Connected components
9. Complexity
10. Dijkstra for weighted graphs

Motivation

From graph data to graph questions

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We learned how to store a graph: vertices, edges, adjacency lists, matrices, directions, and weights.

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- What is the shortest number of steps?

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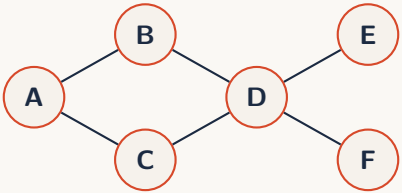
- How many disconnected groups exist?
- What changes when edges have weights?

The core operation is traversal: start somewhere, move through edges, and remember what you have seen.

One graph, many questions



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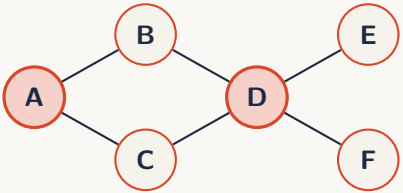

Reachability

Can we get from A to D?

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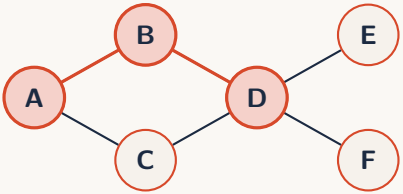

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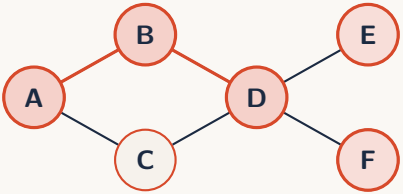


Reachability

Can we get from A to D?

Shortest path

What is the fewest number of edges?

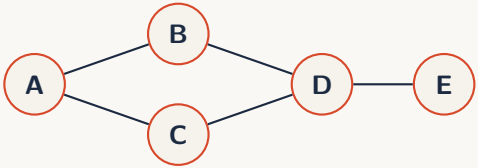
Expansion

What can we discover next?

Recap adjacency lists

Adjacency list recap

```
graph = {
    "A": ["B", "C"],
    "B": ["A", "D"],
    "C": ["A", "D"],
    "D": ["B", "C", "E"],
    "E": ["D"],
}
```



Adjacency list recap

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graph = {
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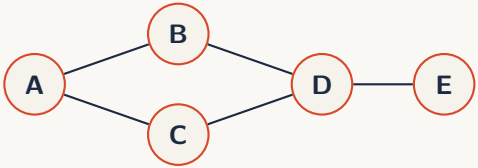

Meaning

`graph[u]` is the list of vertices directly reachable from u.

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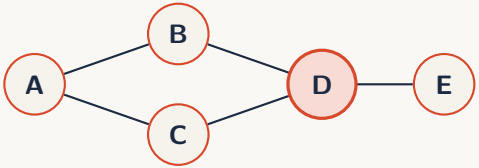

For D, the neighbors are B, C, and E.

Meaning

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The one loop every graph algorithm starts with

```
for neighbor in graph[node]:
    # inspect this edge:
    # node -> neighbor
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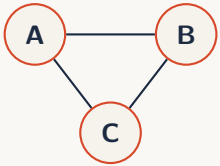

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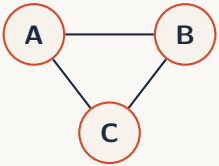

Traversal needs memory

When a graph has cycles, moving blindly is not enough. We need to know which vertices we already visited.

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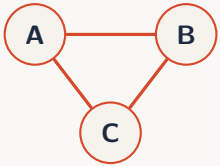

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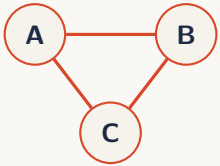

Without `visited`, a triangle can trap the algorithm forever.

Traversal problem

Traversal: visit everything reachable

Definition Graph traversal

A traversal algorithm starts from one vertex and systematically visits vertices reachable through edges.

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- distances;

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Depending on the problem:

- visit order;
- reachable set;
- distances;
- parents;
- components.

The frontier: discovered but not explored

Definition Frontier

During traversal, the frontier contains vertices that have been discovered, but whose neighbors have not yet been explored.

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Different traversal algorithms mainly differ by how they choose the next vertex from the frontier.

Two meaningful ways to choose from the frontier

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Strategy 1: oldest first

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BFS and DFS are not arbitrary algorithms. They are the two simplest fundamental ways to manage the traversal frontier.

BFS

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Breadth-First Search is a traversal strategy that always explores the oldest discovered vertex that has not yet been processed.

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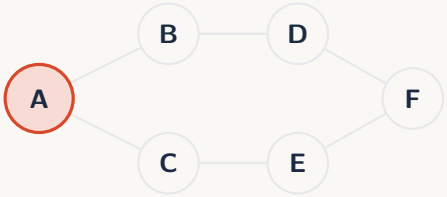
Queue discipline

First discovered, first explored.

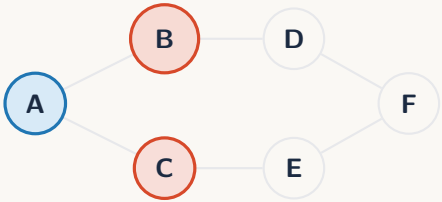
BFS traversal example



Traversal order

A

BFS traversal example



Traversal order

A, B

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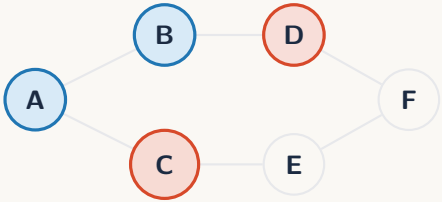
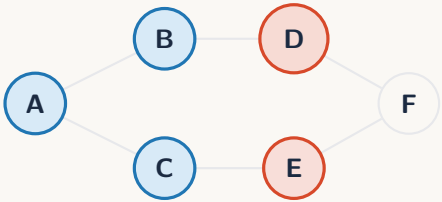


Traversal order

A, B, C

BFS traversal example



Traversal order

A, B, C, D

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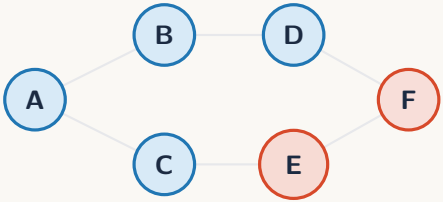
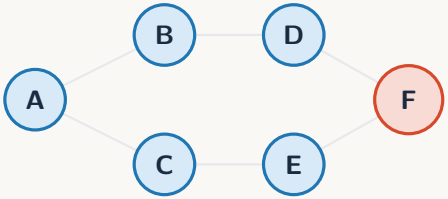


Traversal order

A, B, C, D, E

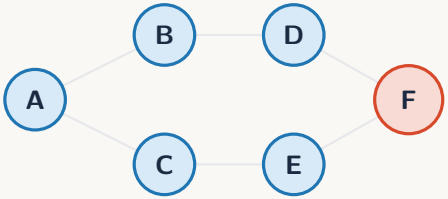
BFS traversal example



Traversal order

A, B, C, D, E, F

BFS traversal example



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Key observation

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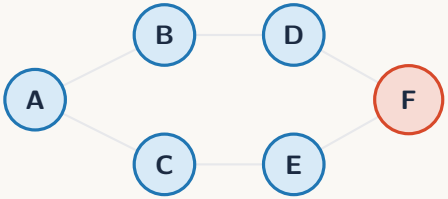

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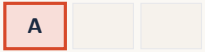
This is the BFS traversal of the graph starting from A.

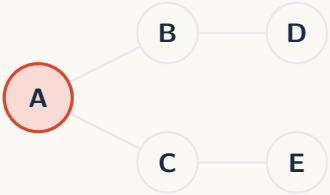
BFS operation: pop from the queue

Current state

Before exploring neighbors, BFS removes the front element of the queue.



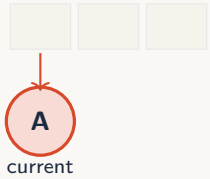
front

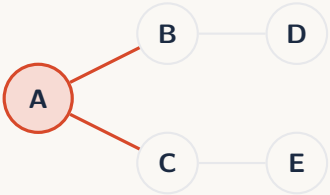


BFS operation: pop from the queue

Current state

Before exploring neighbors, BFS removes the front element of the queue.



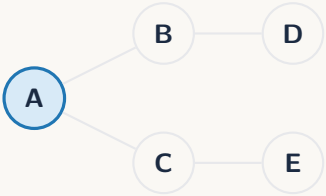


Popping selects the next vertex to process. It does not yet discover anything new.

BFS operation: insert unseen neighbors

After processing A

A has two unseen neighbors: B and C.



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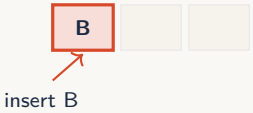

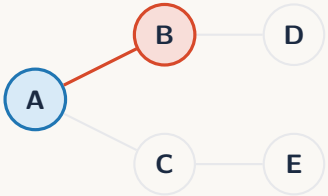

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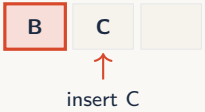
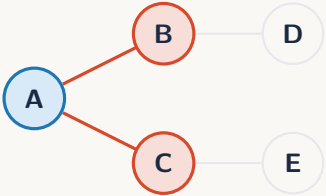


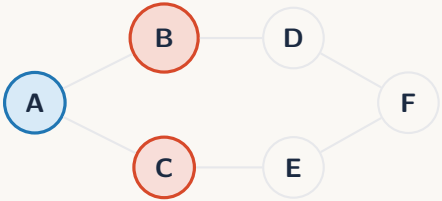


Inserted vertices are discovered, but they are not explored immediately. They wait their turn.

BFS operation: repeat the same rule

Queue before next step

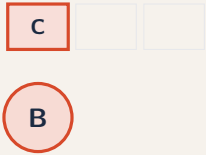
B	C	

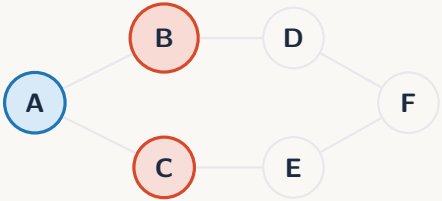


BFS operation: repeat the same rule

Queue before next step

Pop B





BFS operation: repeat the same rule

Queue before next step

B C

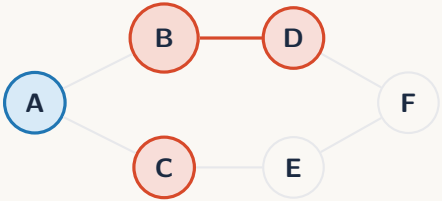
Pop B

C

B

Insert B's unseen neighbor

C D



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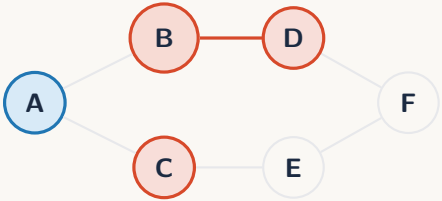
Queue before next step

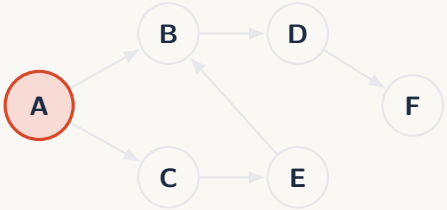
Pop B

Insert B's unseen neighbor



Same rule every time: pop one vertex, inspect its neighbors, insert only unseen ones.

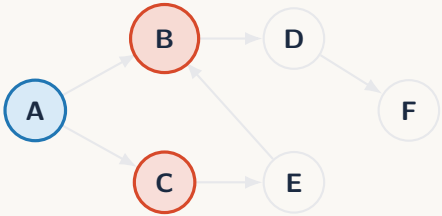
BFS on a directed graph



Directed rule

From a vertex, BFS only follows outgoing edges.

BFS on a directed graph



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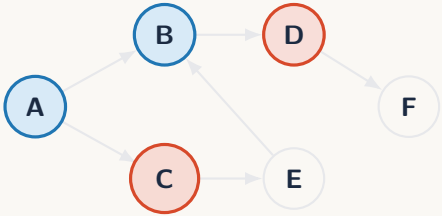
From a vertex, BFS only follows outgoing edges.

Traversal order

A, B

BFS on a directed graph



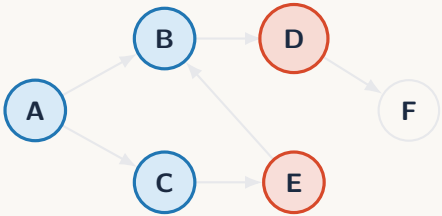
Directed rule

From a vertex, BFS only follows outgoing edges.

Traversal order

A, B, C

BFS on a directed graph



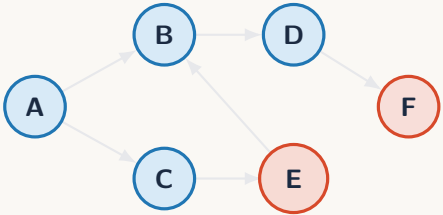
Directed rule

From a vertex, BFS only follows outgoing edges.

Traversal order

A, B, C, D

BFS on a directed graph



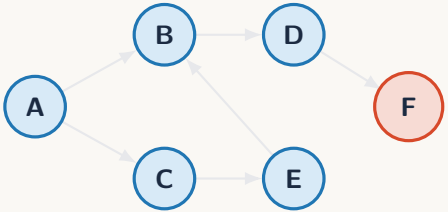
Directed rule

From a vertex, BFS only follows outgoing edges.

Traversal order

A, B, C, D, E

BFS on a directed graph



Directed rule

From a vertex, BFS only follows outgoing edges.

Traversal order

A, B, C, D, E, F

The same queue rule works. Only the meaning of "neighbor" changes: in a directed graph, neighbors are outgoing neighbors.

BFS code

```
1  from collections import deque
2
3  def bfs(graph, start):
4      visited = {start}
5      queue = deque([start])
6
7      while queue:
8          node = queue.popleft()
9          print(node)
10
11         for neighbor in graph[node]:
12             if neighbor not in visited:
13                 visited.add(neighbor)
14                 queue.append(neighbor)
```

BFS code

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The code follows exactly the animation: pop one vertex, inspect its neighbors, insert unseen neighbors.

Shortest path intuition

Now BFS has an important consequence

What we already know

BFS explores vertices using a queue: oldest discovered vertex first.

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BFS explores vertices using a queue: oldest discovered vertex first.

Consequence in unweighted graphs

Because all edges cost one step, this queue order processes vertices by increasing number of edges from the start.

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What we already know

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Consequence in unweighted graphs

Because all edges cost one step, this queue order processes vertices by increasing number of edges from the start.

So BFS is first a traversal algorithm. Shortest paths are a consequence of that traversal order.

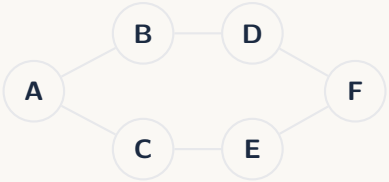
BFS distances in unweighted graphs

```python
def bfs_distances(graph, start):
    distance = {start: 0}
    queue = deque([start])

    while queue:
        node = queue.popleft()

        for neighbor in graph[node]:
            if neighbor not in distance:
                distance[neighbor] = distance[node] + 1
                queue.append(neighbor)

    return distance
```



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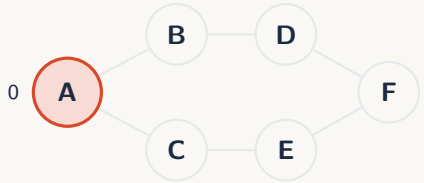

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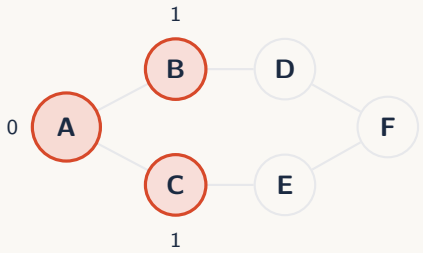

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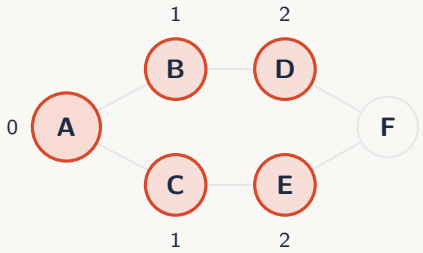

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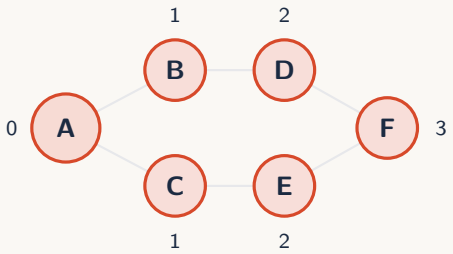

The first time BFS discovers a vertex, it has found the minimum number of edges from the start.

Path reconstruction

Distances are not enough: store parents

```
def shortest_path(graph, start, target):
    parent = {start: None}
    queue = deque([start])

    while queue:
        node = queue.popleft()
        if node == target:
            break

        for neighbor in graph[node]:
            if neighbor not in parent:
                parent[neighbor] = node
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Parent meaning

```python
parent[neighbor] = node
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means: we reached `neighbor` from `node`.

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Parent meaning

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The parent dictionary stores a shortest-path tree rooted at the start vertex.

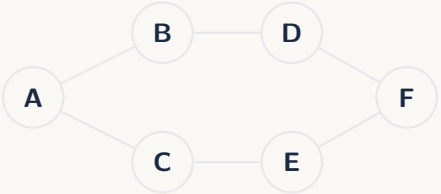
Reconstruct the path backward

```
if target not in parent:
    return None

path = []
current = target

while current is not None:
    path.append(current)
    current = parent[current]

path.reverse()
return path
```



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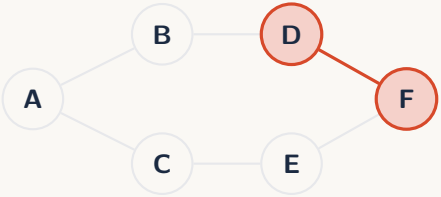

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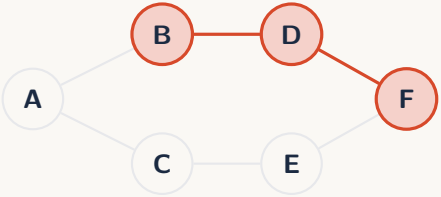

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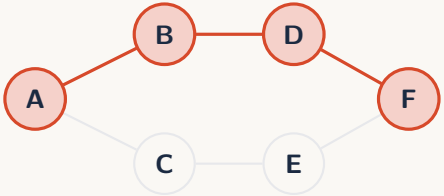
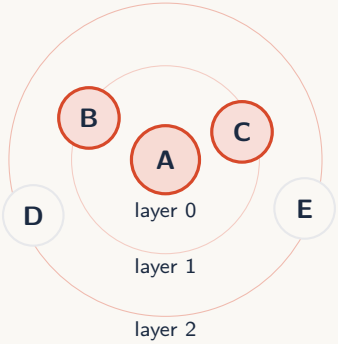


Backward then reverse

$F \leftarrow D \leftarrow B \leftarrow A$ becomes $A \rightarrow B \rightarrow D \rightarrow F$.

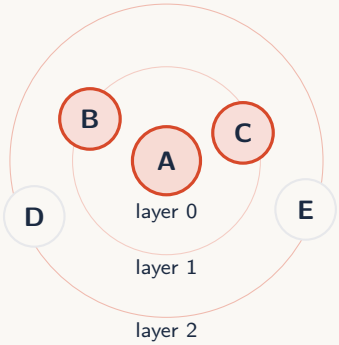
Why BFS gives shortest paths here



Why BFS gives shortest paths here

Layer invariant

Before BFS starts processing distance $k+1$, it has already processed every vertex at distance k.



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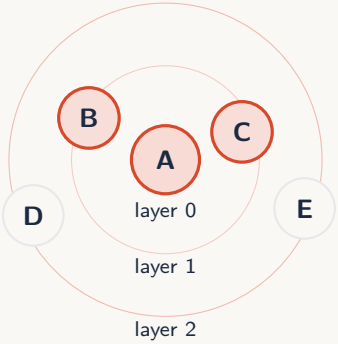
If a target appears for the first time at layer k, no path with fewer than k edges exists.



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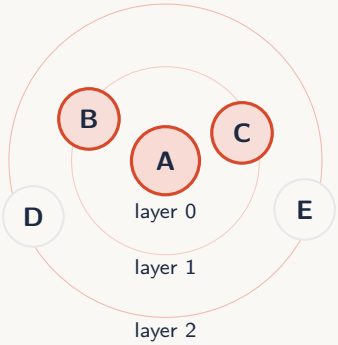

This is only about number of edges. It assumes every edge has equal cost.

DFS

DFS: newest discovered vertex first

Definition Depth-First Search

Depth-First Search is a traversal strategy that always explores the newest discovered vertex that has not yet been processed.

DFS: newest discovered vertex first

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Frontier rule

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- pop from the top;
- insert new unseen neighbors on top;

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Frontier rule

- discovered vertices wait in a stack;
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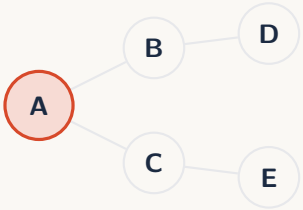
Stack discipline

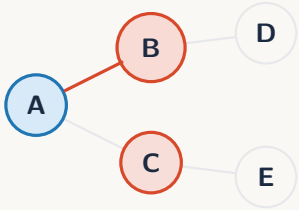
Last discovered, first explored.

DFS traversal example



Traversal order
A

DFS traversal example



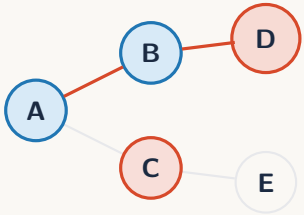
Traversal order

A, B

Key observation

DFS commits to one branch before returning to older alternatives.

DFS traversal example



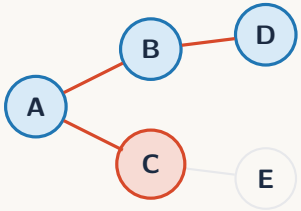
Traversal order

A, B, D

Key observation

DFS commits to one branch before returning to older alternatives.

DFS traversal example



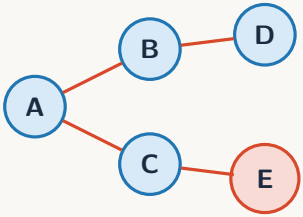
Traversal order

A, B, D, C

Key observation

DFS commits to one branch before returning to older alternatives.

DFS traversal example



Traversal order

A, B, D, C, E

Key observation

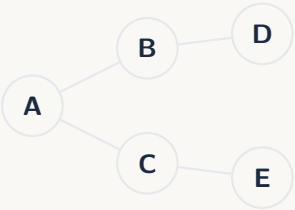
DFS commits to one branch before returning to older alternatives.

This is the DFS traversal of the graph starting from A.

DFS operation: push the start vertex

Initial state

DFS starts by putting the start vertex on the stack.

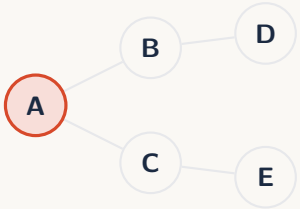


DFS operation: push the start vertex

Initial state

DFS starts by putting the start vertex on the stack.

top



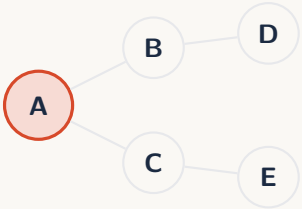
The stack contains vertices that have been discovered but not yet processed.

DFS operation: pop from the stack

Current stack

The top of the stack is the next vertex to process.

top



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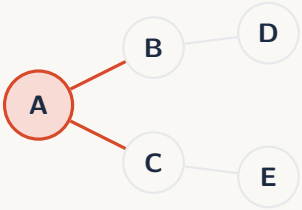
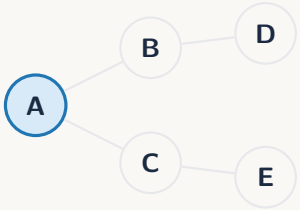


Popping chooses the next vertex. After that, we inspect its neighbors.

DFS operation: push unseen neighbors

After processing A

A has two unseen neighbors: B and C.



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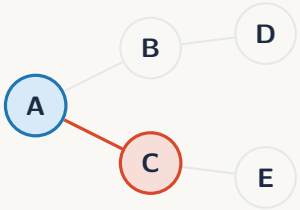

DFS operation: push unseen neighbors

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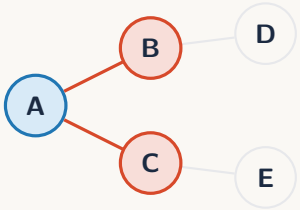


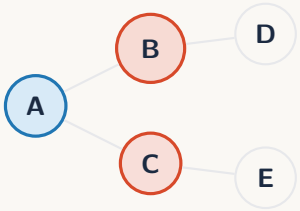
To visit B before C with a stack, we push C first, then B. The last pushed vertex is popped first.

DFS operation: repeat the same rule

Stack before next step

top

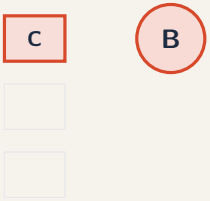
C

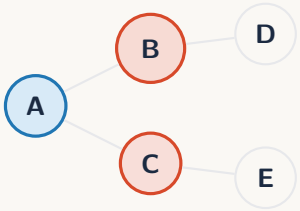


DFS operation: repeat the same rule

Stack before next step

Pop B





DFS operation: repeat the same rule

Stack before next step

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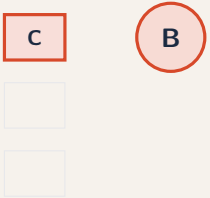
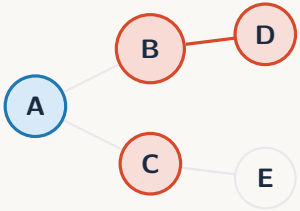


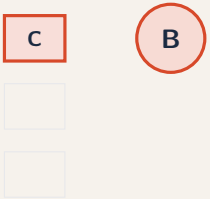


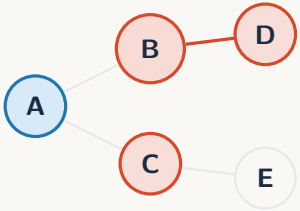
Push B's unseen neighbor

DFS operation: repeat the same rule

Stack before next step

Pop B





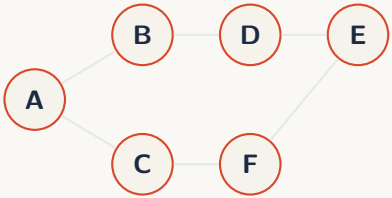
Push B's unseen neighbor

The stack makes DFS continue deeper before returning to C.

DFS is not organized by distance

DFS behavior

DFS follows the newest available vertex.



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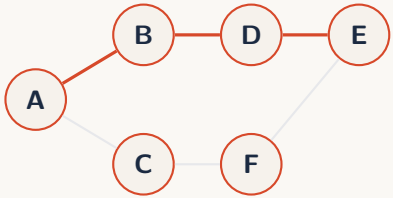
DFS follows the newest available vertex.

Consequence

It may reach a far vertex before a closer one.



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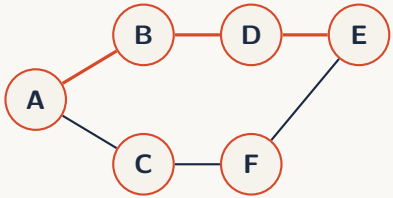

Point

DFS may find one path to E, but not necessarily the shortest one first.

DFS is not organized by distance

DFS behavior

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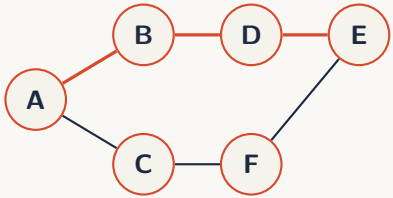
Consequence

It may reach a far vertex before a closer one.

DFS is useful for exploration and structure, but it does not guarantee shortest paths.



Point

DFS may find one path to E, but not necessarily the shortest one first.

DFS code: recursive and iterative

Recursive DFS

```
def dfs(graph, node, visited):
    visited.add(node)
    print(node)

    for neighbor in graph[node]:
        if neighbor not in visited:
            dfs(graph, neighbor,
    visited)
```

Iterative DFS

```
def dfs_iterative(graph, start):
    visited = set()
    stack = [start]
    while stack:
        node = stack.pop()
        if node in visited:
            continue
        visited.add(node)
        print(node)
        for neighbor in
    reversed(graph[node]):
            if neighbor not in
    visited:
                stack.append(neighbor)
```

DFS code: recursive and iterative

Recursive DFS

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```

Recursive DFS is short, but on very deep graphs it can hit Python's recursion limit. Iterative DFS avoids that

BFS vs DFS

Question	BFS	DFS
Data structure	Queue	Stack / recursion
Exploration style	Layer by layer	Deep then backtrack
Unweighted shortest path	Yes	No
Memory behavior	Stores frontier	Stores current branch / stack
Common use	distances, levels	exhaustive search, components

BFS vs DFS

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Memory behavior	Stores frontier	Stores current branch / stack
Common use	distances, levels	exhaustive search, components

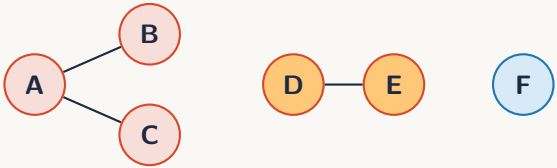
For this course, use BFS when distance matters. Use DFS when reachability or full exploration is enough.

Connected components

Connected components in undirected graphs

Definition Connected component

In an undirected graph, a connected component is a maximal group of vertices where every vertex can reach every other vertex in the group.



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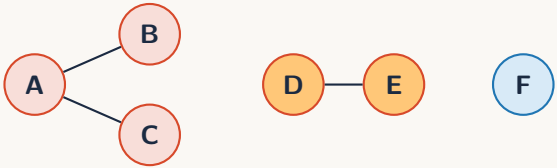

This graph has three connected components: $\{A, B, C\}$, $\{D, E\}$, and $\{F\}$.

Finding components: repeated traversal

```python
def collect_component(graph, start, visited):
    component = []
    stack = [start]
    while stack:
        node = stack.pop()
        if node in visited:
            continue
        visited.add(node)
        component.append(node)
        for neighbor in graph[node]:
            stack.append(neighbor)
    return component
```

```python
def connected_components(graph):
    visited = set()
    components = []
    for node in graph:
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            )

    components.append(component)
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```
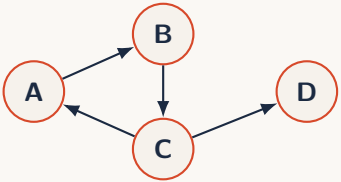

For this course, connected components means undirected connected components unless stated otherwise.

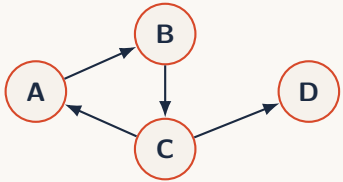
Directed graphs: weak vs strong components



Directed graphs: weak vs strong components

Weakly connected

Ignore arrow directions. If the underlying undirected graph is connected, the directed graph is weakly connected.



Directed graphs: weak vs strong components

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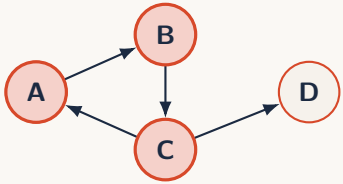
Strongly connected

Respect arrow directions. Every vertex must be able to reach every other vertex.



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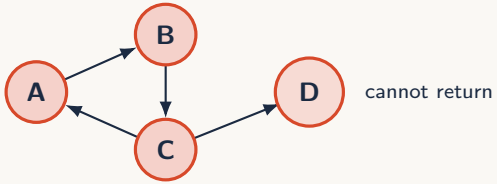
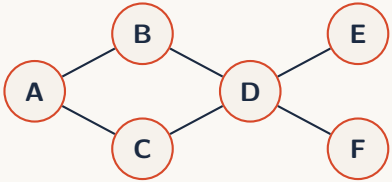


A, B, C form a strongly connected component. D is reached from them, but cannot reach them back.

Complexity

Why BFS and DFS are $O(V + E)$

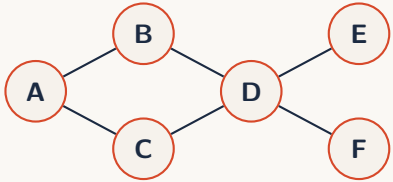


Why BFS and DFS are $O(V + E)$

Vertices

Each vertex is marked visited once.

$$O(V)$$



Why BFS and DFS are $O(V + E)$

Vertices

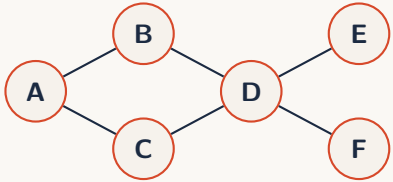
Each vertex is marked visited once.

$$O(V)$$

Edges

Across the whole algorithm, adjacency lists are scanned once per visited vertex.

$$O(E)$$



Complexity summary

Algorithm	Time	Space	Main extra data
BFS traversal	$O(V+E)$	$O(V)$	queue, visited
BFS distances	$O(V+E)$	$O(V)$	queue, distance
BFS path	$O(V+E)$	$O(V)$	queue, parent
DFS traversal	$O(V+E)$	$O(V)$	stack/recursion, visited
Components	$O(V+E)$	$O(V)$	visited, component list

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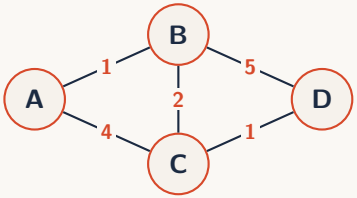
Components	$O(V + E)$	$O(V)$	visited, component list

For adjacency matrices, scanning neighbors costs more: checking all possible neighbors of a node costs $O(V)$.

Dijkstra for weighted graphs

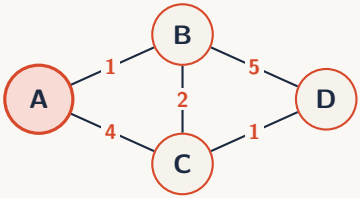
Weighted shortest path: Dijkstra's idea



Weighted shortest path: Dijkstra's idea

Problem

Find the minimum total cost from a start vertex to every other vertex in a graph with non-negative edge weights.



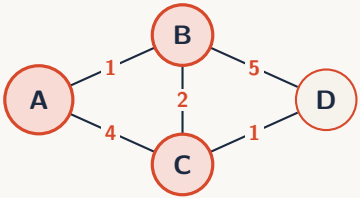
Weighted shortest path: Dijkstra's idea

Problem

Find the minimum total cost from a start vertex to every other vertex in a graph with non-negative edge weights.

Main idea

Always expand the unsettled vertex with the smallest known distance so far.



Weighted shortest path: Dijkstra's idea

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Find the minimum total cost from a start vertex to every other vertex in a graph with non-negative edge weights.

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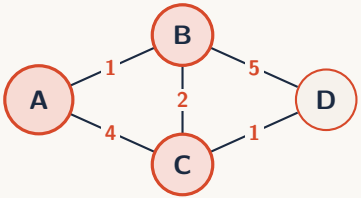
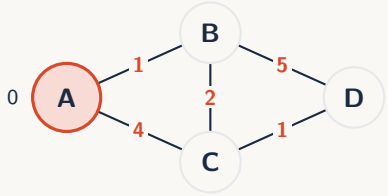
Dijkstra is BFS with a priority queue instead of a normal queue.

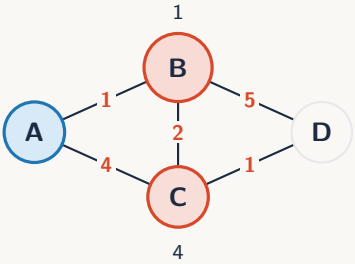


Dijkstra animation: choose smallest tentative distance



Priority queue contents
(0,A)

Dijkstra animation: choose smallest tentative distance



Priority queue contents

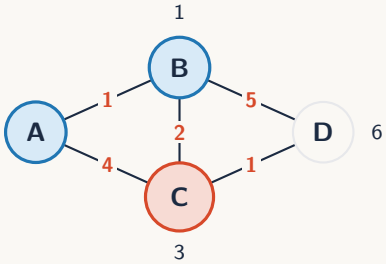
(1,B) (4,C)

Relaxation

When going through edge $u \to v$ with weight w, try:

$$\texttt{dist[v]} = \min(\texttt{dist[v]}, \texttt{dist[u]} + w)$$

Dijkstra animation: choose smallest tentative distance



Priority queue contents

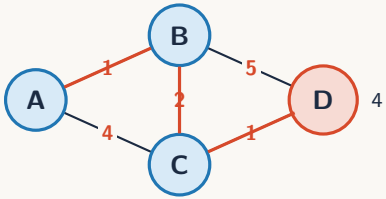
(3,C) (6,D)

Relaxation

When going through edge $u \to v$ with weight w, try:

$$\texttt{dist[v]} = \min(\texttt{dist[v]}, \texttt{dist[u]} + w)$$

Dijkstra animation: choose smallest tentative distance



Priority queue contents

(4,D)

Relaxation

When going through edge $u \to v$ with weight w, try:

$$\texttt{dist[v]} = \min(\texttt{dist[v]}, \texttt{dist[u]} + w)$$

Dijkstra code skeleton

```python
import heapq

def dijkstra(graph, start):
    distance = {start: 0}
    pq = [(0, start)]

    while pq:
        current_dist, node = heapq.heappop(pq)
        if current_dist != distance[node]:
            continue
        for neighbor, weight in graph[node]:
            new_dist = current_dist + weight

            if neighbor not in distance or new_dist < distance[neighbor]:
                distance[neighbor] = new_dist
                heapq.heappush(pq, (new_dist, neighbor))

    return distance
```

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BFS uses a normal queue because all priorities are effectively equal by layer.

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Course rule of thumb: unweighted shortest path means BFS. Weighted shortest path with non-negative weights means Dijkstra.

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4. DFS explores deeply using recursion or a stack.
5. Connected components come from repeated traversal in undirected graphs.
6. Dijkstra extends shortest-path intuition to non-negative weighted graphs using a priority queue.